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Predefined-time cluster lag synchronization of inertial neural networks: A dynamic event-triggered control

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ABSTRACT

This work addresses the problem of predefined-time cluster lag synchronization for inertial neural networks. A dynamic event-triggered control scheme that incorporates a time-dependent exponential scaling function is proposed. Sufficient conditions are established to ensure the achievement of cluster lag synchronization within a prespecified time. In contrast to traditional Lyapunov–Krasovskii functional approaches that usually result in high-dimensional linear matrix inequalities, the criteria obtained in this paper are formulated as low-dimensional linear matrix inequalities that align with the dimension of the systems, which facilitates easy verification. Furthermore, an additional constraint is derived to preclude the occurrence of Zeno behavior in the closed-loop system. Finally, numerical simulation results are presented to validate the effectiveness of the proposed control strategy and the correctness of the theoretical derivations

1. Introduction

Neural networks (NNs) are computational models that mimic the structure of the human brain, characterized by interconnected neurons and synapses. Their advantages lie in parallel processing, adaptability, and the ability to handle nonlinear problems. Inertial neural networks (INNs) are a variant of NNs that more accurately describes the inertial behavior of electronic components in neural network circuits by introducing inertial terms. This generalization enriches the network dynamics and encompasses traditional NNs as a special case when the inertial effects are absent. Due to the incorporation of extra inertial terms in differential equations, INNs exhibit more intricate dynamical behaviors (Dharani et al., 2017; Ge & Xu, 2013; Liu et al., 2022; Song et al., 2020), allowing for a more precise depiction of real-world phenomena. Recently, INNs have been widely used in the fields of intelligent transportation, pattern recognition, natural language processing, etc. Han et al. (2023), a) Li (Zhao), Li et al. (2017), Liu et al. (2023c), c) Tian (Yang). Given that understanding the dynamic behavior of INNs is crucial for their successful applications in various fields, the dynamic behavior analysis of INNs has become a research hotspot, including projection synchronization, bipartite synchronization, cluster synchronization, and leader-following synchronization (Chen et al., 2023, 2020; Hu et al., 2015; Liu et al., 2023b; Udhayakumar et al., 2023). Cluster syn-

chronization means that nodes within the same cluster are synchronized. Considering the communication delay among agents caused by the limitation of transmission speed, cluster lag synchronization has become a recent research hotspot (Cai & Zhou, 2019; Qin et al., 2023; Yang et al., 2019).

Given that coupled systems may not be able to realize synchronization through their own information exchange, a variety of valuable controls have been designed, including pinning control (Mwanandiye et al., 2020; Wang et al., 2015), event-triggered control (Tang et al., 2024; Yu & Antsaklis, 2013), adaptive control (Chen & Lewis, 2010; Kong et al., 2023), and more. Among these, event-triggered control merely relies on triggering conditions to be activated, which reduces the transmission of redundant information and diminishes the number of control while ensuring the system performance, thus greatly saving communication resources. Recently, significant attention is being paid to the synchronization of INNs under event-triggered control (Fu et al., 2026; b) Lu, Li; Shanmugasundaram et al., 2022; Yao et al., 2023; Zhang et al., 2023). For instance, impulsive synchronization with time delays was studied in Shanmugasundaram et al. (2022), exponential synchronization under parameter disturbances was investigated in Yao et al. (2023), and non-reduced order synchronization was addressed in Zhang et al. (2023). It is worth noting that all these event-triggered controls are static in nature, where the threshold for triggering depends on the instantaneous system

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state. Considering that the trigger threshold may decrease over time, which in turn can cause unnecessary event triggering in some cases, some effective controllers such as dynamic event triggering controllers have been proposed to address this issue (Cao & Cao, 2024; Hu et al., 2020; Nie et al., 2022; d)Zhao, Zhao).

In many scenarios, the convergence speed of synchronization is considered an important performance metric. Accordingly, various types of time-constrained synchronization-such as finite-time synchronization (Alimi et al., 2019; Chaouki & Abed, 2021), fixed-time synchronization (Guo et al., 2021; Kong et al., 2021), and predefined-time synchronization (Han et al., 2022, 2024; Yan et al., 2024; Zhang & Cao, 2023)-have attracted considerable attention. Notably, unlike its finite-time and fixed-time counterparts, predefined-time synchronization does not rely on initial conditions or system parameters for its settling time, making it configurable for specific engineering requirements. In Han et al. (2022, 2024), Yan et al. (2024), Zhang and Cao (2023), the predefined-time synchronization of INNs was explored by designing controllers with state-dependent sign functions. In Wang et al. (2018), employed power functions as scaling mechanisms to construct effective controllers. The designed controller therein successfully avoids the internal oscillations of the system that may be caused by controllers with sign functions. Thereafter, this approach inspired subsequent variants using power-function-based scalars for predefined-time synchronization design, e.g., Assali (2021), Ni et al. (2019), Sahoo et al. (2024). In Liu et al. (2023a), exponential functions have also been introduced as scaling functions to formulate control strategies. Given the efficacy of exponential scalars, developing event-triggered controllers using such functions to achieve predefined-time cluster lag synchronization in INNs holds substantial theoretical and practical value.

Building on the preceding analysis, this paper focuses on the predefined-time cluster lag synchronization of INNs. A dynamic event-triggered control strategy is proposed, and sufficient criteria are established to ensure the predefined-time cluster lag synchronization under this framework. The remainder of the paper is structured as follows: Section 2 introduces the fundamental concepts regarding graph theory and describes the system model. In Section 3, a dynamic event-triggered controller is developed for INNs, leading to the derivation of several sufficient synchronization conditions, and the exclusion of Zeno behavior is demonstrated. Section 4 provides numerical simulations to validate the obtained results, while Section 5 concludes the paper and suggests potential future research directions.

2. Preliminaries and model description

Throughout the paper, let $\mathbb{R}$, $\mathbb{Z}^+$, and $\mathbb{R}^{P \times n}$ denote the sets of real numbers, positive integers, and $P \times n$ -dimensional matrices, respectively. The symbol $\otimes$ represents the Kronecker product, while $\hat{K} > 0$ stands for the Euclidean norm. Define I_n as the diagonal matrix with elements $\{1, \dots, 1\} \in \mathbb{R}^{n \times n}$, 0_n as the vector $(0, \dots, 0)^T \in \mathbb{R}^n$, and $O_n := \text{diag}\{0, \dots, 0\} \in \mathbb{R}^{n \times n}$. $\lambda_{\max}(\hat{K})$ represent maximum eigenvalues of $\hat{K} > 0$.

2.1. Graph theory

The interaction topology among nodes is modeled as a directed graph $\mathcal{G} = \{\Delta, W, M\}$ where $\Delta = \{1, \dots, P\}$ is the node set, $W \subseteq \Delta \times \Delta$ constitutes the edge set, and $M = (m_{ij})_{P \times P} \in \mathbb{R}^{P \times P}$ is the adjacency matrix of $\mathcal{G}$. If j links to i , then $m_{ij} > 0$, otherwise $m_{ij} = 0$. The Laplacian matrix associated with $\mathcal{G}$ is given by $L = B - M$, where $B = \text{diag}\{B_1, \dots, B_P\}$ represents in-degree matrix with $B_i = \sum_{j \in \mathcal{N}_i} m_{ij}$ ($\mathcal{N}_i = \{j \in \Delta : (j, i) \in W\}$). Assume that the digraph $\mathcal{G}$ can be partitioned into q ($\Delta_1, \Delta_2, \dots, \Delta_q$) clusters, where $\Delta_1 = \{1, \dots, P_1\}$, $\Delta_2 = \{P_1 + 1, \dots, P_2\}$, ..., $\Delta_q = \{P_{q-1} + 1, \dots, P_q\}$ ($P_q = P$). $\mathcal{G}_{\tilde{i}}$ is the corresponding subgraph with regard to $\tilde{i}$ -th clusters, and $\mathcal{P}_{\tilde{i}} = P_i - P_{i-1}$ ($\tilde{i} = 1, \dots, q$) indicates the number of nodes within the $\tilde{i}$ -th cluster. It should be noted that $\mathcal{P}_{\tilde{i}}$ may differ from $\mathcal{P}_{\tilde{j}}$ for

$\tilde{i} \neq \tilde{j}$, meaning that different clusters can contain different numbers of nodes.

Assumption 1. For $\tilde{i}, \tilde{j} = 1, \dots, q$, L can be formulated as

$$L = \begin{bmatrix} L_{11} & \dots & L_{1q} \\ \vdots & \ddots & \vdots \\ L_{q1} & \dots & L_{qq} \end{bmatrix},$$

where $L_{\tilde{i}\tilde{j}} \in \mathbb{R}^{P_{\tilde{i}} \times P_{\tilde{j}}}$ ($\tilde{i} \neq \tilde{j}$) has zero row-sums, $L_{\tilde{i}\tilde{i}} \in \mathbb{R}^{P_{\tilde{i}} \times P_{\tilde{i}}}$ denotes a matrix with zero row sums and negative diagonal elements.

Assumption 1 specifies that interactions within the same cluster are cooperative (positive), while interactions between different clusters may be either cooperative or competitive (negative). This assumption ensures the stability of the state of cluster synchronization. Specifically, the condition that $L_{\tilde{i}\tilde{j}} \in \mathbb{R}^{P_{\tilde{i}} \times P_{\tilde{j}}}$ ($\tilde{i} \neq \tilde{j}$) has zero row sums implies that each cluster remains unaffected by others after achieving cluster synchronization, thereby maintaining the synchronized state.

2.2. Model description

Consider a system comprising P INNs ($P > 0$). The dynamics of the i -th ($i = 1, \dots, P$) INN is presented as:

$$\frac{d^2 \zeta_i(t)}{dt^2} = -A \frac{d\zeta_i(t)}{dt} - Z\zeta_i(t) + Fg(\zeta_i(t)) + I_v + r_i(t), \quad (1)$$

where $\zeta_i(t) = (\zeta_{i1}(t), \dots, \zeta_{in}(t))^T$ is the state vector. $\frac{d^2 \zeta_i(t)}{dt^2}$ is the inertia term. $A = \text{diag}\{a_1, \dots, a_n\}$ and $Z = \text{diag}\{z_1, \dots, z_n\}$ are the self-feedback positive definite matrix. $I_v = (I_1, \dots, I_n)^T \in \mathbb{R}^n$ stands for external input vector. $F = (f_{ij})_{n \times n}$ is referred to as the matrix of connection weights. $r_i(t) \in \mathbb{R}^n$ denotes the control input. $g(\zeta_i(t)) = (g_1(\zeta_i(t)), \dots, g_n(\zeta_i(t)))^T$ indicates the vector-valued of continuous activation functions satisfying the Assumption 2.

Assumption 2. For $\forall \alpha_1, \alpha_2 \in \mathbb{R}^n$, there exists $\ell > 0$ such that

$$\|g(\alpha_1) - g(\alpha_2)\| \leq \ell \|\alpha_1 - \alpha_2\|.$$

The initial conditions for the INNs (1) are provided as: for $i = 1, \dots, P$,

$$\begin{cases} \zeta_i(0) = \varrho_i, \\ \frac{d\zeta_i(0)}{dt} = \tilde{\varrho}_i, \end{cases} \quad (2)$$

where $\varrho_i, \tilde{\varrho}_i \in \mathbb{R}$.

Define the following isolated INNs:

$$\frac{d^2 S_{\tilde{i}}(t)}{dt^2} = -A \frac{dS_{\tilde{i}}(t)}{dt} - ZS_{\tilde{i}}(t) + Fg(S_{\tilde{i}}(t)) + I_v, \quad (3)$$

where $S_{\tilde{i}}(t) = (S_{\tilde{i}1}(t), \dots, S_{\tilde{i}n}(t))^T$.

Definition 1. The INNs (1) are said to realize predefined-time cluster lag synchronization within a predefined time T_p if for $\tilde{i}, \tilde{j} = 1, \dots, q$,

$$\lim_{t \rightarrow T_p} \|\zeta_i(t) - S_{\tilde{i}}(t - \tau_i)\| = 0, \quad \forall i \in \Delta_{\tilde{i}}$$

and

$$\|\zeta_i(t) - S_{\tilde{i}}(t - \tau_i)\| \equiv 0, \quad \forall t \geq T_p, \quad \forall i \in \Delta_{\tilde{i}}.$$

In Definition 1, the cluster lag synchronization condition only requires INNs within the same cluster to synchronize to their corresponding virtual leader. For INNs belonging to different clusters, their synchronized trajectories are not required to coincide, and they may evolve independently. Note that τ_i represents the lag time of the nodes in the $\tilde{i}$ -th cluster lagging behind their corresponding leaders $S_{\tilde{i}}(t)$, $1 \leq \tilde{i} \leq q$. In this paper, we use $S_{\tilde{i}}(t - \tau_i)$ to represent $S_{\tilde{i}}(t - \tau_i)$ without causing ambiguity.

To streamline the analysis, the variable transformation method is adopted: for $\tilde{i} = 1, \dots, q$,

$$h_i(t) = \frac{d\zeta_i(t)}{dt} + \kappa \zeta_i(t), \quad i \in \Delta_{\tilde{i}}$$

and

$$\bar{H}_i(t) = \frac{dS_i(t)}{dt} + \kappa S_i(t),$$

where $0 \leq \kappa \leq 1$, then the INNs (1) and the isolated INNs (3) can be depicted as:

$$\begin{cases} \frac{d\zeta_i(t)}{dt} = -\kappa\zeta_i(t) + \hat{h}_i(t), \\ \frac{d\hat{h}_i(t)}{dt} = (\kappa A - \kappa^2 I_n - Z)\zeta_i(t) + (\kappa I_n - A)\hat{h}_i(t) \\ + Fg(\zeta_i(t)) + I_v + r_i(t), \end{cases} \quad (4)$$

and

$$\begin{cases} \frac{dS_i(t)}{dt} = -\kappa S_i(t) + \bar{H}_i(t), \\ \frac{d\bar{H}_i(t)}{dt} = (\kappa A - \kappa^2 I_n - Z)S_i(t) + Fg(S_i(t)) \\ + (\kappa I_n - A)\bar{H}_i(t) + I_v. \end{cases} \quad (5)$$

Let $s_i(t) = \zeta_i(t) - S_i(t_{\tau_i}) \in \mathbb{R}^n$, $h_i(t) = \hat{h}_i(t) - \bar{H}_i(t_{\tau_i}) \in \mathbb{R}^n$ with $i \in \Delta_i$ and $t_{\tau_i} = t - \tau_i > 0$. According to (4) and (5), one gets

$$\begin{cases} \frac{ds_i(t)}{dt} = -\kappa s_i(t) + h_i(t), \\ \frac{dh_i(t)}{dt} = (\kappa A - \kappa^2 I_n - Z)s_i(t) + (\kappa I_n - A)h_i(t) \\ + Fg(s_i(t)) + r_i(t), \end{cases}$$

where $g(s_i(t)) = g(\zeta_i(t)) - g(S_i(t_{\tau_i})) \in \mathbb{R}^n$.

The above equation can be reformulated into an equivalent expression:

$$\begin{bmatrix} \dot{s}_i(t) \\ \dot{h}_i(t) \end{bmatrix} = \begin{bmatrix} -\kappa I_n & I_n \\ \kappa A - \kappa^2 I_n - Z & \kappa I_n - A \end{bmatrix} \begin{bmatrix} s_i(t) \\ h_i(t) \end{bmatrix} + \begin{bmatrix} 0_n \\ Fg(s_i(t)) \end{bmatrix} + \begin{bmatrix} 0_n \\ r_i(t) \end{bmatrix}. \quad (6)$$

Denote $\varpi(t) = (s^T(t), h^T(t))^T \in \mathbb{R}^{2Pn}$ with $s(t) = (s_1^T(t), \dots, s_P^T(t))^T \in \mathbb{R}^{Pn}$ and $h(t) = (h_1^T(t), \dots, h_P^T(t))^T \in \mathbb{R}^{Pn}$, $r(t) = (r_1^T(t), \dots, r_P^T(t))^T \in \mathbb{R}^{Pn}$. From (6) one gets

$$\dot{\varpi}(t) = \begin{bmatrix} -\kappa I_{Pn} & I_{Pn} \\ \Xi_1 & \Xi_2 \end{bmatrix} \varpi(t) + \begin{bmatrix} 0_{Pn} \\ F(t) \end{bmatrix} + \begin{bmatrix} 0_{Pn} \\ r(t) \end{bmatrix}, \quad (7)$$

where $\Xi_1 = I_P \otimes (\kappa A - \kappa^2 I_n - Z)$, $\Xi_2 = I_P \otimes (\kappa I_n - A)$, and $F(t) = (Fg(s_1(t)), \dots, Fg(s_P(t))) \in \mathbb{R}^{Pn}$.

From the definition of $\varpi(t)$, we have the following equivalence: the INNs (1) realize predefined-time cluster lag synchronization if and only if $\lim_{t \rightarrow T_p^-} \|\varpi(t)\| = 0$ and $\|\varpi(t)\| = 0$ for $t \geq T_p$.

3. Main results

This section presents a dynamic event-triggered control scheme. Sufficient conditions are established to ensure the predefined-time cluster lag synchronization of INNs, and Zeno behavior is analysed to confirm its elimination.

3.1. Control design

Define the following time-dependent scaling function:

$$\mathcal{J}(t) = \begin{cases} \frac{e^{\mu T_p} - 1}{e^{\mu(T_p - t)} - 1}, & t \in [0, T_p), \\ e^{\mu(t - T_p)}, & t \in [T_p, +\infty), \end{cases} \quad (8)$$

where $T_p > 0$ and $\mu > 0$ represent the predefined time and constants, respectively. From (8) we have

$$\dot{\mathcal{J}}(t) = \begin{cases} \mu \mathcal{J}(t)(1 + \frac{\mathcal{J}(t)}{e^{\mu T_p} - 1}), & t \in [0, T_p), \\ \mu e^{\mu(t - T_p)}, & t \in [T_p, +\infty). \end{cases} \quad (9)$$

Denote $d\zeta_i(t)/dt = \hat{\zeta}_i(t)$ and $dS_i(t_{\tau_i})/dt = \hat{S}_i(t_{\tau_i})$, then define the following event-triggered function: for $i \in \Delta_i$,

$$\varphi_i(t) = \sum_{j=1}^P \omega_{ij} m_{ij} (\zeta_j(t) - \zeta_i(t) + \hat{\zeta}_j(t) - \hat{\zeta}_i(t)) + \omega_i d_i (\hat{S}_i(t_{\tau_i})$$

$$- \hat{\zeta}_i(t) + S_i(t_{\tau_i}) - \zeta_i(t)), \quad (10)$$

where ω_{ij} signifies the intercluster coupling strength and is given by $\omega_{ij} = \omega_i > 0$ if $i, j \in \Delta_i$ and $\omega_{ij} = 1$ if $i, j \notin \Delta_i$. $M = (m_{ij})_{P \times P}$ represents the adjacency matrix, where $m_{ij} > 0$ for $i \neq j$ and $m_{ii} = 0$. For each cluster $\bar{i} = 1, \dots, q$, an augmented graph $\bar{\mathcal{G}}_{\bar{i}}$ is constructed by adding the virtual leader node $S_{\bar{i}}(t)$ to the cluster subgraph $\mathcal{G}_{\bar{i}}$. The leader node is connected to a subset of nodes within the cluster, referred to as pinned nodes. Denote by $Y_{\bar{i}} \subseteq \Delta_{\bar{i}}$ the set of pinned nodes in the $\bar{i}$ -th cluster. For each pinned node $i \in Y_{\bar{i}}$, the pinning gain satisfies $d_i > 0$, which corresponds to the direct control input from the leader in the controller design.

Denote a new graph $\bar{\mathcal{G}}_{\bar{i}} (\bar{i} = 1, \dots, q)$, which is formed by the graph $\mathcal{G}_{\bar{i}}$ of the $\bar{i}$ -th cluster together with its associated leader node $S_{\bar{i}}(t)$. Denote by $Y_{\bar{i}}$ the set of pinned nodes selected from the $\bar{i}$ -th cluster. This indicates that if $i \in Y_{\bar{i}}$, then $d_i > 0$. if $i \notin Y_{\bar{i}}$, then $d_i = 0$.

Assumption 3. $\bar{\mathcal{G}}_{\bar{i}}$ contains a directed spanning tree ($\bar{i} = 1, \dots, q$).

Denote

$$\bar{L} = \begin{bmatrix} \omega_1 \bar{L}_{11} & L_{12} & \dots & L_{1q} \\ L_{21} & \omega_2 \bar{L}_{22} & \dots & L_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ L_{q1} & L_{q2} & \dots & \omega_q \bar{L}_{qq} \end{bmatrix},$$

where $\bar{L}_{\bar{i}\bar{i}} = L_{\bar{i}\bar{i}} + D_{\bar{i}}$, $D = \text{diag}\{d_1, \dots, d_P\} = \text{diag}\{D_1, \dots, D_q\}$, and $D_{\bar{i}} \in \mathbb{R}^{P \times P_{\bar{i}}} (\bar{i} = 1, \dots, q)$. We state the following lemma:

Lemma 1. *Qin et al. (2019)* Given *Assumptions 1 and 3*, one can find positive diagonal matrices $\Lambda_{\bar{i}} > 0$ ($\bar{i} = 1, \dots, q$) satisfying $\Lambda_{\bar{i}} \bar{L}_{\bar{i}\bar{i}} + \bar{L}_{\bar{i}\bar{i}}^T \Lambda_{\bar{i}} > 0$, or equivalently, there exists $\rho > 0$ satisfying $\Lambda \bar{L} + \bar{L}^T \Lambda > \rho \Lambda$, where $\Lambda = \text{diag}\{\Lambda_1, \dots, \Lambda_q\}$.

Based on the definition of scaling function (8) (10), introduce the following control:

$$r_i(t) = \frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)} \varphi_i(t_i^l), \quad t \in [t_i^l, t_{i+1}^l). \quad (11)$$

For $i = 1, \dots, P$, define the measurement error of node i as:

$$\begin{cases} \delta_i(t) = \zeta_i(t_i^l) - \zeta_i(t) - (S_i(t_{\tau_i}) - S_i(t_{\tau_i}^l)), \\ \bar{\delta}_i(t) = \hat{h}_i(t_i^l) - \hat{h}_i(t) - (\bar{H}_i(t_{\tau_i}) - \bar{H}_i(t_{\tau_i}^l)), \end{cases}$$

where $i \in \Delta_i$, $t \in [t_i^l, t_{i+1}^l)$, $S_i(t_{\tau_i})$ and $\bar{H}_i(t_{\tau_i})$ are the transmission state of leader node at $t_{\tau_i} = t_i^l - \tau_i$. t_i^l stands for the l -th triggering instant of the i -th nodes with $t_0^l = 0$. The dynamic event-triggering condition takes the following form:

$$\begin{aligned} t_1^l &= 0, \\ t_{l+1}^l &= \inf\{t : t > t_l^l, \|\psi_i(t)\| > \sigma \|\varphi_i(t_l^l)\| + \varepsilon_i(t)\}, \end{aligned} \quad (12)$$

where $0 < \sigma < \min\{1, \frac{\|\bar{L}\|}{\sqrt{3}}\}$, $\psi_i(t) = (\delta_i^T(t), \bar{\delta}_i^T(t))^T$, and $\varepsilon_i(t)$ is continuous and defined by

$$\dot{\varepsilon}_i(t) = -(\zeta_1 + \zeta_2 \frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)}) \varepsilon_i(t) + \sigma \|\varphi_i(t_l^l)\| - \|\psi_i(t)\| \quad (13)$$

with $\varepsilon_i(0), \zeta_1, \zeta_2 > 0$. The function $\varepsilon_i(t)$ enables adaptive adjustment of the triggering threshold. In particular, its dynamics can temporarily increase the threshold during the initial control phase, thereby reducing the frequency of triggering events, while the overall mechanism also helps prevent Zeno behavior.

Lemma 2. For $i = 1, \dots, P$, $\varepsilon_i(t) > 0$ for $t \in [0, T_p)$ and $\varepsilon_i(t) \geq 0$ for $t \in [T_p, +\infty)$.

Proof. We first prove $\varepsilon_i(t) > 0$ for $t \in [0, T_p)$. For a fixed i , let $\{t_l^l\}_{l=1}^\infty$ be the triggering instants on $[0, T_p)$ with $t_1^l = 0$. On each interval (t_l^l, t_{l+1}^l) , the dynamic event-triggered mechanism yields the differential inequality

$$\dot{\varepsilon}_i(t) \geq -\left(\zeta_1 + 1 + \zeta_2 \frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)}\right) \varepsilon_i(t), \quad (14)$$

where $\varsigma_1, \varsigma_2 > 0$ are constants. From the definition of $\mathcal{J}(t)$ in (8), for $t \in [0, T_p)$ we have

$$\frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)} = \mu \frac{e^{\mu(T_p-t)}}{e^{\mu(T_p-t)} - 1} = \mu \left(1 + \frac{1}{e^{\mu(T_p-t)} - 1} \right).$$

Substituting this into (14) gives

$$\begin{aligned} \dot{\varepsilon}_i(t) &\geq -\left(\varsigma_1 + 1 + \varsigma_2\mu + \frac{\varsigma_2\mu}{e^{\mu(T_p-t)} - 1}\right)\varepsilon_i(t) \\ &= -\left(\bar{\varsigma}_1 + \frac{\bar{\varsigma}_2}{e^{\mu(T_p-t)} - 1}\right)\varepsilon_i(t), \end{aligned} \quad (15)$$

where $\bar{\varsigma}_1 = \varsigma_1 + 1 + \varsigma_2\mu$ and $\bar{\varsigma}_2 = \varsigma_2\mu$ are positive constants. Define $g(t) := \bar{\varsigma}_1 + \frac{\bar{\varsigma}_2}{e^{\mu(T_p-t)} - 1}$, which is continuous on $[0, T_p)$ and satisfies

$$\int_0^t g(s) ds = \bar{\varsigma}_1 t + \frac{\bar{\varsigma}_2}{\mu} \ln \frac{e^{\mu T_p} - 1}{e^{\mu(T_p-t)} - 1} < \infty \quad \text{for any } t < T_p.$$

Applying the comparison lemma to (15) on $(t_i^i, t_{i+1}^i]$ yields

$$\varepsilon_i(t) \geq \varepsilon_i(t_i^i) \exp\left(-\int_{t_i^i}^t g(s) ds\right), \quad t \in (t_i^i, t_{i+1}^i].$$

Since $\varepsilon_i(0) > 0$ and $t_1^i = 0$, iterating over all intervals gives

$$\varepsilon_i(t) \geq \varepsilon_i(0) \exp\left(-\int_0^t g(s) ds\right) > 0, \quad \forall t \in [0, T_p).$$

Thus $\varepsilon_i(t) > 0$ for all $t \in [0, T_p)$. In particular, the right-hand limit $\varepsilon_i(T_p^-) := \lim_{t \rightarrow T_p^-} \varepsilon_i(t)$ exists and satisfies $\varepsilon_i(T_p^-) \geq 0$.

Now consider $t \geq T_p$. For $t > T_p$, the scaling function $\mathcal{J}(t) = e^{\mu(t-T_p)}$ gives $\frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)} = \mu$. Substituting into (14) yields

$$\dot{\varepsilon}_i(t) \geq -(\varsigma_1 + 1 + \varsigma_2\mu)\varepsilon_i(t) \triangleq -\gamma\varepsilon_i(t), \quad t > T_p,$$

with $\gamma > 0$. Applying the comparison lemma again, for any $t > T_p$,

$$\varepsilon_i(t) \geq \varepsilon_i(T_p^+)e^{-\gamma(t-T_p)},$$

where $\varepsilon_i(T_p^+)$ denotes the right-hand limit at $t = T_p$. Since $\varepsilon_i(t)$ is continuous, we have $\varepsilon_i(T_p^+) = \varepsilon_i(T_p^-) \geq 0$. Consequently, $\varepsilon_i(t) \geq 0$ for all $t \geq T_p$. This completes the proof. $\square$

Let $\psi(t) = (\delta^T(t), \bar{\delta}^T(t))^T \in \mathbb{R}^{2Pn}$ with $\delta(t) = (\delta_1^T(t), \dots, \delta_p^T(t))^T \in \mathbb{R}^{Pn}$, $\bar{\delta}(t) = (\bar{\delta}_1^T(t), \dots, \bar{\delta}_p^T(t))^T \in \mathbb{R}^{Pn}$, the control (11) admits the compact representation:

$$r(t) = \frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)} \begin{bmatrix} O_{Pn} & O_{Pn} \\ -(1-\kappa)\bar{L} \otimes I_n & -\bar{L} \otimes I_n \end{bmatrix} (\varpi(t) + \psi(t)). \quad (16)$$

Then, substituting (16) into (7) yields to

$$\begin{aligned} \dot{\varpi}(t) &= \begin{bmatrix} -\kappa I_{Pn} & I_{Pn} \\ \Xi_1 - \frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)}(1-\kappa)\bar{L} \otimes I_n & \Xi_2 - \frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)}\bar{L} \otimes I_n \end{bmatrix} \varpi(t) \\ &+ \begin{bmatrix} O_{Pn} \\ F(t) \end{bmatrix} - \frac{\dot{\mathcal{J}}(t)}{\mathcal{J}(t)} \begin{bmatrix} O_{Pn} & O_{Pn} \\ (1-\kappa)\bar{L} \otimes I_n & \bar{L} \otimes I_n \end{bmatrix} \psi(t). \end{aligned} \quad (17)$$

3.2. Predefined-time cluster lag synchronization analysis

Denote $L_s = L - \text{diag}\{L_{11}, \dots, L_{qq}\}$, $\xi_1 = \kappa^2 + (3\kappa - 2\kappa^2 - 1)\rho - 2\kappa^3$, $\xi_2 = -2\kappa + 2\kappa^2 - |2\kappa - 1|$, $\xi_3 = -(1-\kappa)\rho - \kappa^2 - 2$, $\xi_4 = 2 - |2\kappa - 1|$, and $\xi_5 = \kappa^2 + (3\kappa - 2\kappa^2 - 1 + \ell)\rho - 2\kappa^3 - 2\ell$, where ρ is defined in Lemma 1. Besides, $w_4 = \min\{w_1, 2\varsigma_2 - w_2\}$, where $w_1 = \min\left\{\lambda_{\min}\left(\frac{1}{4}\Lambda \otimes (2\kappa^2 - 3\kappa + 1)\rho I_n\right), \lambda_{\min}\left(\frac{1}{4}\Lambda \otimes (\kappa + 1)\rho I_n\right)\right\} - \frac{3\theta_1\sigma^2\|\bar{L}\|^2}{1-3\sigma^2\|\bar{L}\|^2}$, $w_2 = \frac{3\theta_1}{1-3\sigma^2\|\bar{L}\|^2}$. The following results can be obtained.

Theorem 1. The INNs (1) can realize predefined-time cluster lag synchronization under Assumptions 1–3 if

$$\omega_i > \max\left\{0, \frac{-\lambda_{\min}(\Lambda L_s + L_s^T \Lambda)}{\lambda_{\min}(\Lambda_i \bar{L}_{ii} + \bar{L}_{ii}^T \Lambda_i)}\right\}, \quad (18)$$

$$\begin{cases} \xi_5 I_n + \xi_2 A - (2\kappa - 1)Z - (1-\rho)FF^T > 0, \\ \xi_3 I_n + \xi_4 A - Z - FF^T > 0, \end{cases} \quad (19)$$

and $T_p > \max\{\tau_i\}$ for $i = 1, \dots, q$, $\rho > 1 - \kappa$, $1 - \frac{w_4}{2\epsilon} < 0$. Meanwhile, $r_i(t)$ is bounded for $t \geq 0$.

Proof. Consider the Lyapunov function candidate:

$$V(t) = V_1(t) + V_2(t),$$

where $V_1(t) = \varpi^T(t)Y\varpi(t)$ and $V_2(t) = \sum_{i=1}^P \varepsilon_i^2(t)$ with $Y = \begin{bmatrix} (1-\kappa)\hat{L} & (1-\kappa)\Lambda \\ (1-\kappa)\Lambda & \Lambda \end{bmatrix} \otimes I_n$ and $\hat{L} = \Lambda\bar{L} + \bar{L}^T\Lambda$.

We claim that Y is positive definite. In fact, according to Schur's lemma, $Y > 0$ is equivalent to $\Lambda > 0$ and

$$S = (1-\kappa)\hat{L} - [(1-\kappa)\Lambda](\Lambda^{-1})[(1-\kappa)\Lambda] = (1-\kappa)\hat{L} - (1-\kappa)^2\Lambda > 0.$$

According to Lemma 1, we have $(1-\kappa)\hat{L} \geq \rho(1-\kappa)\Lambda$ and $\Lambda > 0$, which incorporation with $0 < \kappa < 1$ shows $S > 0$, hence Y is positive definite.

For $t \in [0, T_p)$, according to (17), we have

$$\begin{aligned} \dot{V}_1(t) &= 2\varpi^T(t)Y \begin{bmatrix} -\kappa I_{Pn} & I_{Pn} \\ \Xi_1 & \Xi_2 \end{bmatrix} \varpi(t) + 2\varpi^T(t)Y \begin{bmatrix} O_{Pn} \\ F(t) \end{bmatrix} \\ &- \frac{2\dot{\mathcal{J}}(t)}{\mathcal{J}(t)} \varpi^T(t)Y \begin{bmatrix} O_{Pn} & O_{Pn} \\ (1-\kappa)\bar{L} \otimes I_n & \bar{L} \otimes I_n \end{bmatrix} (\varpi(t) + \psi(t)). \end{aligned}$$

Base on the definition of Y ,

$$\begin{aligned} 2Y &= \begin{bmatrix} -\kappa I_{Pn} & I_{Pn} \\ \Xi_1 & \Xi_2 \end{bmatrix} \\ &= \begin{bmatrix} (1-\kappa)\hat{L} & (1-\kappa)\Lambda \\ (1-\kappa)\Lambda & \Lambda \end{bmatrix} \otimes I_n \begin{bmatrix} -\kappa I_{Pn} & I_{Pn} \\ \Xi_1 & \Xi_2 \end{bmatrix} \begin{bmatrix} \eta_1 & \eta_2 \\ \eta_2^T & \eta_3 \end{bmatrix}, \end{aligned}$$

where $\eta_1 = 2(\Lambda \otimes ((\kappa^3 - \kappa^2)I_n + (\kappa - \kappa^2)A - (1-\kappa)Z) - (\kappa - \kappa^2)\hat{L} \otimes I_n)$, $\eta_2 = \Lambda \otimes ((2\kappa - 1)A - \kappa^2 I_n - Z) - (1-\kappa)\hat{L} \otimes I_n$, and $\eta_3 = 2\Lambda \otimes (I_n - A)$. Let

$$R_1(t) = 2\varpi^T(t)Y \begin{bmatrix} -\kappa I_{Pn} & I_{Pn} \\ \Xi_1 & \Xi_2 \end{bmatrix} \varpi(t),$$

then

$$R_1(t) = \begin{bmatrix} s^T(t) \\ h^T(t) \end{bmatrix} \begin{bmatrix} \eta_1 & \eta_2 \\ \eta_2^T & \eta_3 \end{bmatrix} \begin{bmatrix} s(t) \\ h(t) \end{bmatrix} \quad (20)$$

In accordance with Lemma 1 and Young's Inequality, one gets

$$\begin{aligned} &s^T(t)\eta_1 s(t) \\ &= 2s^T(t) \left(\Lambda \otimes ((\kappa^3 - \kappa^2)I_n + (\kappa - \kappa^2)A - (1-\kappa)Z) \right. \\ &\quad \left. - (\kappa - \kappa^2)\hat{L} \otimes I_n \right) s(t) \\ &\leq 2s^T(t) \left(\Lambda \otimes ((\kappa^3 - \kappa^2 - (\kappa - \kappa^2)\rho)I_n + (\kappa - \kappa^2)A \right. \\ &\quad \left. - (1-\kappa)Z) \right) s(t) \end{aligned} \quad (21)$$

and

$$\begin{aligned} &s^T(t)\eta_2 h(t) + h^T(t)\eta_2^T s(t) \\ &= 2s^T(t) \left(\Lambda \otimes ((2\kappa - 1)A - \kappa^2 I_n - Z) - (1-\kappa)\hat{L} \otimes I_n \right) h(t) \\ &\leq s^T(t) \left(\Lambda \otimes ((1-\kappa)\rho + |2\kappa - 1|A + \kappa^2)I_n + Z \right) s(t) \\ &\quad + h^T(t) \left(\Lambda \otimes (|2\kappa - 1|A + ((1-\kappa)\rho + \kappa^2)I_n + Z) \right) h(t). \end{aligned} \quad (22)$$

From above arguments, one can obtain

$$\begin{aligned} R_1(t) &\leq s^T(t) \left(\Lambda \otimes \left(2((\kappa^3 - \kappa^2 - (\kappa - \kappa^2)\rho)I_n + (\kappa - \kappa^2)A \right. \right. \\ &\quad \left. \left. - (1-\kappa)Z) \right) + (|2\kappa - 1|A + ((1-\kappa)\rho + \kappa^2)I_n + Z) \right) s(t) \end{aligned}$$

$$\begin{aligned}
& + h^T(t) \left(\Lambda \otimes \left((|2\kappa - 1|A + ((1 - \kappa)\rho + \kappa^2)I_n + Z) \right. \right. \\
& \left. \left. + 2(I_n - A) \right) \right) h(t) \\
& \leq -s^T(t) \left(\Lambda \otimes (\xi_1 I_n + \xi_2 A - (2\kappa - 1)Z) \right) s(t) \\
& \quad - h^T(t) (\Lambda \otimes (\xi_3 I_n + \xi_4 A - Z)) h(t).
\end{aligned} \tag{23}$$

In light of [Assumption 2](#), we have

$$\begin{aligned}
& 2\varpi^T(t)Y \begin{bmatrix} 0_{P_n} \\ F(t) \end{bmatrix} \\
& = 2s^T(t)((1 - \rho)\Lambda \otimes I_n)F(t) + 2h^T(t)(\Lambda \otimes I_n)F(t) \\
& \leq (1 - \rho)s^T(t)(\Lambda \otimes FF^T)s(t) + h^T(t)(\Lambda \otimes FF^T)h(t) \\
& \quad + (2 - \rho)g^T(s(t))(\Lambda \otimes I_n)g(s(t)) \\
& \leq s^T(t) \left(\Lambda \otimes ((1 - \rho)FF^T + (2 - \rho)\mathcal{L}I_n) \right) s(t) \\
& \quad + h^T(t)(\Lambda \otimes FF^T)h(t).
\end{aligned} \tag{24}$$

Denote $R_2(t) = -2\frac{\dot{J}(t)}{J(t)}\varpi^T(t)Y \begin{bmatrix} 0_{P_n} & O_{P_n} \\ (1 - \kappa)\bar{L} \otimes I_n & \bar{L} \otimes I_n \end{bmatrix} \cdot (\varpi(t) + \psi(t))$. According to Young's Inequality, one can derive

$$\begin{aligned}
R_2(t) & \leq -\frac{\dot{J}(t)}{2J(t)} \begin{bmatrix} s^T(t) \\ h^T(t) \end{bmatrix} \begin{bmatrix} \hat{\eta}_1 & \hat{\eta}_2 \\ \hat{\eta}_2^T & \hat{\eta}_3 \end{bmatrix} \begin{bmatrix} s(t) \\ h(t) \end{bmatrix} + \frac{\dot{J}(t)}{J(t)} \\
& \quad \cdot \psi^T(t) \begin{bmatrix} (1 - \kappa)^2 \bar{L}^T \bar{L} \otimes I_n & (1 - \kappa) \bar{L}^T \bar{L} \otimes I_n \\ (1 - \kappa) \bar{L}^T \bar{L} \otimes I_n & \bar{L}^T \bar{L} \otimes I_n \end{bmatrix} \psi(t),
\end{aligned}$$

where $\hat{\eta}_1 = (1 - \kappa)^2 \bar{L} \otimes I_n$, $\hat{\eta}_2 = (1 - \kappa) \bar{L} \otimes I_n$, and $\hat{\eta}_3 = \bar{L} \otimes I_n$. By means of [Lemma 1](#),

$$\begin{aligned}
-s^T(t)\hat{\eta}_1 s(t) & = -s^T(t)((1 - \kappa)^2 \bar{L} \otimes I_n)s(t) \\
& \leq -s^T(t)(\Lambda \otimes (1 - \kappa)^2 \rho I_n)s(t)
\end{aligned} \tag{25}$$

and

$$-h^T(t)\hat{\eta}_3 h(t) = -h^T(t)(\bar{L} \otimes I_n)h(t) \leq -h^T(t)(\Lambda \otimes \rho I_n)h(t). \tag{26}$$

Similarly,

$$\begin{aligned}
& -s^T(t)\hat{\eta}_2 h(t) - h^T(t)\hat{\eta}_2^T s(t) \\
& = -2s^T(t)((1 - \kappa)\bar{L} \otimes I_n)h(t) \leq -s^T(t)(\Lambda \otimes (1 - \kappa)\rho I_n)h(t) \\
& \leq \frac{1 - \kappa}{2} \left(s^T(t)(\Lambda \otimes \rho I_n)s(t) + h^T(t)(\Lambda \otimes \rho I_n)h(t) \right).
\end{aligned} \tag{27}$$

It follows from [\(12\)](#) that

$$\begin{aligned}
\|\psi_i(t)\| & \leq \sigma \|\varphi_i(t_i^i)\| + \varepsilon_i(t) \\
& \leq \sigma \left\| -\sum_{j=1}^P \omega_{ij} l_{ij} ((1 - \kappa)\delta_j(t) + \bar{\delta}_j(t)) - \omega_i d_i ((1 - \kappa)\delta_i(t) \right. \\
& \quad \left. + \bar{\delta}_i(t)) \right\| + \sigma \left\| -\sum_{j=1}^P \omega_{ij} l_{ij} ((1 - \kappa)s_j(t) + h_j(t)) \right. \\
& \quad \left. - \omega_i d_i ((1 - \kappa)s_j(t) + h_j(t)) \right\| + \varepsilon_i(t),
\end{aligned}$$

which can further yield $\|\psi(t)\|^2 \leq 3\sigma^2 \|\bar{L}\|^2 \|\psi(t)\|^2 + 3\sigma^2 \|\bar{L}\|^2 \|\varpi(t)\|^2 + 3 \sum_{i=1}^P \varepsilon_i^2(t)$. Due to the fact that $0 < \sigma < \min\{1, \frac{\|\bar{L}\|}{\sqrt{3}}\}$, we have

$$\|\psi(t)\|^2 \leq \frac{3\sigma^2 \|\bar{L}\|^2}{1 - 3\sigma^2 \|\bar{L}\|^2} \|\varpi(t)\|^2 + \frac{3 \sum_{i=1}^P \varepsilon_i^2(t)}{1 - 3\sigma^2 \|\bar{L}\|^2},$$

which follows that

$$\begin{aligned}
& \psi^T(t) \begin{bmatrix} (1 - \kappa)^2 \bar{L}^T \bar{L} \otimes I_n & (1 - \kappa) \bar{L}^T \bar{L} \otimes I_n \\ (1 - \kappa) \bar{L}^T \bar{L} \otimes I_n & \bar{L}^T \bar{L} \otimes I_n \end{bmatrix} \psi(t) \\
& \leq \psi^T(t) \begin{bmatrix} (2 + \kappa^2 - 3\kappa) \bar{L}^T \bar{L} \otimes I_n & O_{P_n} \\ O_{P_n} & (2 - \kappa) \bar{L}^T \bar{L} \otimes I_n \end{bmatrix} \psi(t)
\end{aligned}$$

$$\leq \frac{3\theta_1 \sigma^2 \|\bar{L}\|^2}{1 - 3\sigma^2 \|\bar{L}\|^2} \|\varpi(t)\|^2 + \frac{3\theta_1 \sum_{i=1}^P \varepsilon_i^2(t)}{1 - 3\sigma^2 \|\bar{L}\|^2}, \tag{28}$$

where $\theta_1 = \lambda_{\max} \begin{bmatrix} (2 + \kappa^2 - 3\kappa) \bar{L}^T \bar{L} \otimes I_n & O_{P_n} \\ O_{P_n} & (2 - \kappa) \bar{L}^T \bar{L} \otimes I_n \end{bmatrix}$.

From [\(25\)-\(28\)](#), we can conclude that

$$\begin{aligned}
R_2(t) & \leq \frac{\dot{J}(t)}{J(t)} \left(-s^T(t) \left(\frac{1}{4} \Lambda \otimes (2\kappa^2 - 3\kappa + 1) \rho I_n \right) s(t) \right. \\
& \quad \left. - h^T(t) \left(\frac{1}{4} \Lambda \otimes (\kappa + 1) \rho I_n \right) h(t) \right. \\
& \quad \left. + \frac{3\theta_1 \sigma^2 \|\bar{L}\|^2}{1 - 3\sigma^2 \|\bar{L}\|^2} \|\varpi(t)\|^2 + \frac{3\theta_1 \sum_{i=1}^P \varepsilon_i^2(t)}{1 - 3\sigma^2 \|\bar{L}\|^2} \right) \\
& \leq -w_1 \frac{\dot{J}(t)}{J(t)} \|\varpi(t)\|^2 + w_2 \frac{\dot{J}(t)}{J(t)} \sum_{i=1}^P \varepsilon_i^2(t),
\end{aligned} \tag{29}$$

where $w_1 = \min\left\{ \lambda_{\min}\left(\frac{1}{4} \Lambda \otimes (2\kappa^2 - 3\kappa + 1) \rho I_n\right), \lambda_{\min}\left(\frac{1}{4} \Lambda \otimes (\kappa + 1) \rho I_n\right) \right\} - \frac{3\theta_1 \sigma^2 \|\bar{L}\|^2}{1 - 3\sigma^2 \|\bar{L}\|^2}$, $w_2 = \frac{3\theta_1}{1 - 3\sigma^2 \|\bar{L}\|^2}$.

Combining with [\(23\)](#), [\(24\)](#), and [\(29\)](#), we have

$$\begin{aligned}
\dot{V}_1(t) & \leq -s^T(t) \left(\Lambda \otimes (\xi_5 I_n + \xi_2 A - (2\kappa - 1)Z - (1 - \rho)FF^T) \right) s(t) \\
& \quad - h^T(t) (\Lambda \otimes (\xi_3 I_n + \xi_4 A - Z - FF^T)) h(t) \\
& \quad - w_1 \frac{\dot{J}(t)}{J(t)} \|\varpi(t)\|^2 + w_2 \frac{\dot{J}(t)}{J(t)} \sum_{i=1}^P \varepsilon_i^2(t) \\
& \leq -\theta_2 \|\varpi(t)\|^2 - w_1 \frac{\dot{J}(t)}{J(t)} \|\varpi(t)\|^2 + w_2 \frac{\dot{J}(t)}{J(t)} \sum_{i=1}^P \varepsilon_i^2(t),
\end{aligned}$$

where $\theta_2 = \min\left\{ \lambda_{\min}\left(\Lambda \otimes (\xi_5 I_n + \xi_2 A - (2\kappa - 1)Z - (1 - \rho)FF^T)\right), \lambda_{\min}\left(\Lambda \otimes (\xi_3 I_n + \xi_4 A - Z - FF^T)\right) \right\}$.

By means of [\(12\)](#) and [\(13\)](#),

$$\begin{aligned}
\dot{V}_2(t) & = -2\varsigma_1 \sum_{i=1}^P \varepsilon_i^2(t) - 2\varsigma_2 \frac{\dot{J}(t)}{J(t)} \sum_{i=1}^P \varepsilon_i^2(t) \\
& \quad + 2 \sum_{i=1}^P \varepsilon_i(t) (\sigma \|\varphi_i(t_i^i)\| - \|\psi_i(t)\|) \\
& \leq -2(\varsigma_1 + 1) \sum_{i=1}^P \varepsilon_i^2(t) - 2\varsigma_2 \frac{\dot{J}(t)}{J(t)} \sum_{i=1}^P \varepsilon_i^2(t).
\end{aligned}$$

From the discussions above, one has

$$\begin{aligned}
\dot{V}(t) & \leq -\theta_2 \|\varpi(t)\|^2 - 2(\varsigma_1 + 1) \sum_{i=1}^P \varepsilon_i^2(t) \\
& \quad - w_1 \frac{\dot{J}(t)}{J(t)} \|\varpi(t)\|^2 - (2\varsigma_2 - w_2) \frac{\dot{J}(t)}{J(t)} \sum_{i=1}^P \varepsilon_i^2(t) \\
& \leq -(w_3 + w_4 \frac{\dot{J}(t)}{J(t)}) (\|\varpi(t)\|^2 + \sum_{i=1}^P \varepsilon_i^2(t)),
\end{aligned}$$

where $w_3 = \min\{\theta_2, 2(\varsigma_1 + 1)\}$, $w_4 = \min\{w_1, 2\varsigma_2 - w_2\}$.

Let $\hat{\lambda} = \lambda_{\max}(Y)$, then we have $V(t) \leq e(\|\varpi(t)\|^2 + \sum_{i=1}^P \varepsilon_i^2(t))$ with $\epsilon = \max\{\hat{\lambda}, 1\}$. Moreover,

$$\dot{V}(t) \leq -\frac{w_3}{\epsilon} V(t) - \frac{w_4}{\epsilon} \frac{\dot{J}(t)}{J(t)} V(t). \tag{30}$$

One can observe from [\(19\)](#) and $1 - \frac{w_4}{2\epsilon} < 0$ that $\frac{w_3}{\epsilon} > 0$ and $\frac{w_4}{\epsilon} > 0$. Therefore, multiplying both sides of inequality [\(30\)](#) by $J(t) \frac{w_4}{\epsilon}$, we get

$$\begin{aligned}
J(t) \frac{w_4}{\epsilon} \dot{V}(t) & \leq -\frac{w_3}{\epsilon} J(t) \frac{w_4}{\epsilon} V(t) - \frac{w_4}{\epsilon} \frac{\dot{J}(t)}{J(t)} J(t) \frac{w_4}{\epsilon} V(t) \\
& \leq -\frac{w_3}{\epsilon} J(t) \frac{w_4}{\epsilon} V(t) - \frac{dJ(t) \frac{w_4}{\epsilon}}{dt} V(t).
\end{aligned}$$

Employing the chain rule of differentiation, we obtain from the inequality that $\frac{dJ(t)}{dt} \leq -\frac{w_3}{\epsilon} J(t) - \frac{w_4}{\epsilon} V(t)$, and consequently,

$$V(t) \leq e^{-\frac{w_3}{\epsilon} t} J(t) - \frac{w_4}{\epsilon} J(0) + \frac{w_4}{\epsilon} V(0). \quad (31)$$

Since $\lim_{t \rightarrow T_p^-} J(t) = 0$ and $J(0) = 1$, we have

$$\lim_{t \rightarrow T_p^-} V(t) \leq \lim_{t \rightarrow T_p^-} e^{-\frac{w_3}{\epsilon} t} J(t) - \frac{w_4}{\epsilon} V(0) = 0.$$

In accordance with the positive definiteness of $V(t)$, we have $\lim_{t \rightarrow T_p^-} V(t) = 0$.

On the other hand, for $t \geq T_p$, from (8) and (9) we have $\frac{J(t)}{J(t)} = \mu$. Substituting this into (30) yields

$$\dot{V}(t) \leq -\left(\frac{w_3}{\epsilon} + \mu \frac{w_4}{\epsilon}\right) V(t), \quad t \in [T_p, +\infty).$$

entailing $V(t) \equiv 0$ for $t \in [T_p, +\infty)$. One may then infer that $\lim_{t \rightarrow T_p^-} \|\varpi(t)\| = 0$ and $\|\varpi(t)\| = 0$ for $t \geq T_p$, implying the achievement of predefined-time cluster lag synchronization for INNs (1).

In the following, we establish the boundedness of the control input $r(t)$.

When $t \in [0, T_p]$, by means of (16),

$$\begin{aligned} \|r(t)\| &\leq \frac{\dot{J}(t)}{J(t)} \|\mathcal{L}(\|\varpi(t)\| + \|\psi(t)\|)\| \\ &\leq \mu \|\mathcal{L}\| (1 + \frac{J(t)}{e^{\mu T_p} - 1}) \left(\frac{\|\varpi(t)\|}{1 - \sigma \|\bar{L}\|} + \frac{\varepsilon_i(t)}{1 - \sigma \|\bar{L}\|} \right) \\ &\leq \frac{\mu \|\mathcal{L}\| (1 + J(t))}{1 - \sigma \|\bar{L}\|} \sqrt{V(t)} \\ &\leq \frac{\mu \|\mathcal{L}\| \sqrt{V(0)}}{1 - \sigma \|\bar{L}\|} e^{-\frac{w_3}{2\epsilon} t} \left(J(t) - \frac{w_4}{2\epsilon} + J(t) 1 - \frac{w_4}{2\epsilon} \right), \end{aligned}$$

where $\mathcal{L} = \begin{bmatrix} O_{P_n} & O_{P_n} \\ -(1 - \kappa) \bar{L} \otimes I_n & -\bar{L} \otimes I_n \end{bmatrix}$. Since $\frac{w_4}{2\epsilon} > 1$ by the conditions of Theorem 1, and $J(t) \geq 1$ on $[0, T_p]$ and $J(t) \rightarrow \infty$ as $t \rightarrow T_p^-$, the right-hand side is uniformly bounded, indicating that $r(t)$ is bounded for $t \in [0, T_p]$.

When $t \in [T_p, +\infty)$, it is easy to verify $\frac{\dot{J}(t)}{J(t)} \equiv \mu$. Thus, $r(t)$ is always bounded for all $t \geq 0$. $\square$

Remark 1. It is worth noting that the order of the linear matrix inequalities (LMIs) in the sufficient conditions of Theorem 1 coincide with the dimensions of the INNs (1). This indicates that converting INNs (1) into high-order systems and analyzing the predefined-time cluster lag synchronization of INNs through the construction of high-order Lyapunov function does not cause the increase in the dimensionality of the LMIs in the sufficiency conditions.

Remark 2. In Alimi et al. (2019), Chaouki and Abed (2021), Chen et al. (2023, 2020), Guo et al. (2021), Hu et al. (2015), Kong et al. (2021), the synchronization of INNs has been investigated, with an emphasis on asymptotic, finite-time, and fixed-time synchronization. In contrast, this paper focuses on predefined-time cluster lag synchronization of INNs, which distinguishes it from the aforementioned works. Moreover, recent studies on predefined-time synchronization of INNs (Han et al., 2022, 2024; Yan et al., 2024; Zhang & Cao, 2023) primarily rely on controllers utilizing state-dependent sign functions. On the contrary, this paper adopts a different approach by incorporating exponential functions as scaling elements in the controller design, which not only avoids chattering but also enables flexible time-varying adjustment that is well suited for the proposed dynamic event-triggered mechanism.

For the special case where the entire network forms a single cluster ($q = 1$), the predefined-time cluster lag synchronization problem reduces to the standard predefined-time lag synchronization for the INNs (1) with respect to a single virtual leader. The corresponding sufficient conditions are given in the following corollary:

Corollary 1. Given Assumptions 2 and 3, the INNs (1) achieve predefined-time lag synchronization if the following conditions hold:

$$\begin{cases} \xi_5 I_n + \xi_2 A - (2\kappa - 1)Z - (1 - \rho)FF^T > 0, \\ \xi_3 I_n + \xi_4 A - Z - FF^T > 0, \end{cases}$$

along with $\rho > 1 - \kappa$, $1 - \frac{w_4}{2\epsilon} < 0$.

When $\tau_i = 0$ ($i = 1, \dots, q$), the predefined-time cluster lag synchronization is equivalent to the predefined-time cluster synchronization. This leads to the corollary:

Corollary 2. The INNs (1) can realize the predefined-time cluster synchronization under Assumptions 2 and 3, if $\rho > 1 - \kappa$, $1 - \frac{w_4}{2\epsilon} < 0$ and inequalities (18) and (19) hold.

3.3. Zeno behavior analysis

Theorem 2. Zeno behavior is excluded when the sufficient conditions provided by Theorem 1 are satisfied for $t < T_p$.

Proof. For $[t_i^j, t_{i+1}^j)$, in view of Assumption 2,

$$\begin{aligned} D^+ \|\psi_i(t)\| &\leq \frac{\|\dot{\psi}_i(t)\|}{\|\psi_i(t)\|} \|\psi_i(t)\| \\ &\leq \left\| \begin{bmatrix} -\kappa I_n & I_n \\ \kappa A - \kappa^2 I_n - Z & \kappa I_n - A \end{bmatrix} \right\| \|\varpi_i(t)\| \\ &\quad + \left\| \begin{bmatrix} O_n & O_n \\ F & O_n \end{bmatrix} \right\| \begin{bmatrix} g(s_i(t)) \\ g(h_i(t)) \end{bmatrix} + \|r_i(t)\| \\ &\leq \chi \|\varpi_i(t)\| + \frac{\dot{J}(t)}{J(t)} \|\varphi_i(t_i^j)\|, \end{aligned} \quad (32)$$

where $\chi = \|X_1\| + \ell \|X_2\|$ with $X_1 = \begin{bmatrix} -\kappa I_n & I_n \\ \kappa A - \kappa^2 I_n - Z & \kappa I_n - A \end{bmatrix}$, $X_2 = \begin{bmatrix} O_n & O_n \\ F & O_n \end{bmatrix}$. Recalling (16), we have

$$\|\varphi_i(t_i^j)\| \leq \Theta_1 (\|\varpi_i(t)\| + \|\psi_i(t)\|),$$

where $\Theta_1 = \left\| \begin{bmatrix} -(1 - \kappa) \sum_{j=1}^P \bar{L}_{ij} & -\sum_{j=1}^P \bar{L}_{ij} \end{bmatrix} \right\|$. According to the dynamic event-triggered strategy (12), one obtains $\|\varphi_i(t_i^j)\| \leq \Theta_1 \|\varpi_i(t)\| + \sigma \Theta_1 \|\varphi_i(t_i^j)\| + \Theta_1 \varepsilon_i(t)$. If $1 - \sigma \Theta_1 > 0$, we have

$$\|\varphi_i(t_i^j)\| \leq \frac{\Theta_1}{1 - \sigma \Theta_1} \|\varpi_i(t)\| + \frac{\Theta_1}{1 - \sigma \Theta_1} \varepsilon_i(t). \quad (33)$$

Let $\bar{\lambda} = \lambda_{\min}(\Upsilon)$, we have $\|\varpi_i(t)\| \leq \sqrt{\frac{1}{\bar{\lambda}}} V_1(t)$, which combined with (33) yields

$$\begin{aligned} D^+ \|\psi_i(t)\| &\leq \left(\chi + \frac{\Theta_1 \dot{J}(t)}{(1 - \sigma \Theta_1) J(t)} \right) \sqrt{\frac{V_1(t)}{\bar{\lambda}}} + \frac{\Theta_1 \dot{J}(t)}{(1 - \sigma \Theta_1) J(t)} \varepsilon_i(t) \\ &\leq \chi \sqrt{V(t)} + 2\Theta_2 \frac{\dot{J}(t)}{J(t)} \sqrt{V(t)}, \end{aligned}$$

where $\Theta_2 = \frac{\Theta_1}{1 - \sigma \Theta_1}$. According to (31), it is easy to see that

$$\begin{aligned} D^+ \|\psi_i(t)\| &\leq \chi e^{-\frac{w_3}{2\epsilon} t} J(t) - \frac{w_4}{2\epsilon} \sqrt{V(0)} + 2\Theta_2 \frac{\dot{J}(t)}{J(t)} e^{-\frac{w_3}{2\epsilon} t} J(t) - \frac{w_4}{2\epsilon} \sqrt{V(0)} \\ &\leq \hat{\Gamma} + \hat{\Gamma} \left(1 + \frac{J(t)}{e^{\mu T_p} - 1} \right) J(t) - \frac{w_4}{2\epsilon} \\ &\leq \hat{\Gamma} + \hat{\Gamma} J(t_i^j) 1 - \frac{w_4}{2\epsilon} \leq \Gamma \end{aligned} \quad (34)$$

holds for $1 - \frac{w_4}{2\epsilon} < 0$ and $\Gamma > 0$, where $\hat{\Gamma} = \chi e^{-\frac{w_3}{2\epsilon} t_i^j} J(t_i^j) - \frac{w_4}{2\epsilon} \sqrt{V(0)}$ and $\hat{\Gamma} = 2\Theta_2 e^{-\frac{w_3}{2\epsilon} t_i^j} \sqrt{V(0)}$. By virtue of (34), one has

$$\|\psi_i(t)\| \leq \int_{t_i^j}^t D^+ \|\psi_i(s)\| ds \leq \int_{t_i^j}^t \Gamma ds \leq \Gamma(t - t_i^j).$$

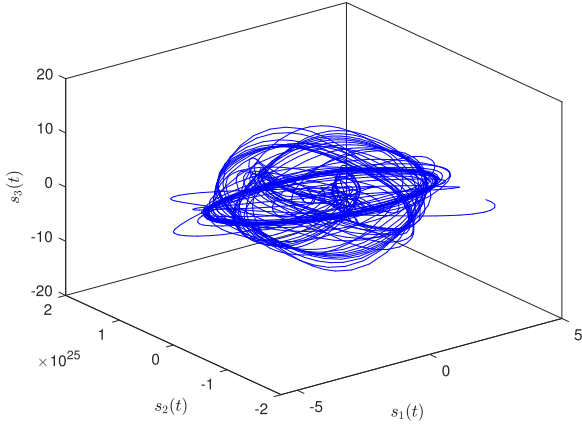


Fig. 1. The portraits of INNs (3) with $S_1(0) = (0.4, 0.7, 0.36)^T$.

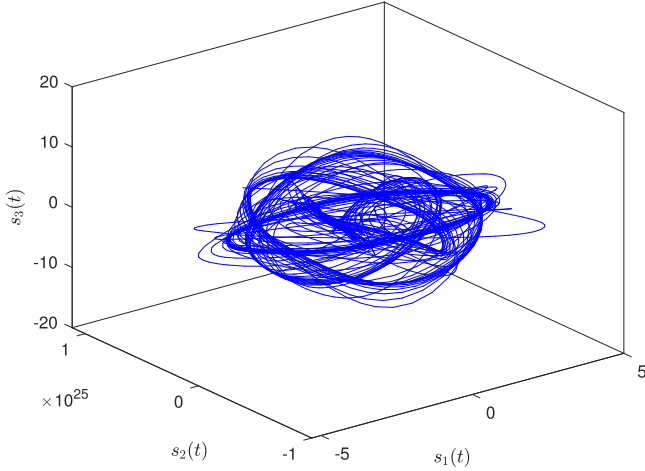


Fig. 2. The portraits of INNs (3) with $S_2(0) = (-0.2, -0.3, 0.14)^T$.

Based on the dynamic event-triggered strategy (12), we derive that $\|\psi_i(t)\| = \sigma\|\varphi_i(t_l^i)\| + \varepsilon_i(t)$ at $t = t_{l+1}^i - 0$. Then we have

$$\sigma\|\varphi_i(t_l^i)\| + \varepsilon_i(t_{l+1}^i) \leq \Gamma T_l, \quad (35)$$

where $T_l = t_{l+1}^i - t_l^i$. For $t \geq 0$, we have $\sigma\|\varphi_i(t_l^i)\| + \varepsilon_i(t) > 0$ based on Lemma 2, which implies that $T_l > 0$. We claim that Zeno behavior can be eliminated during the control period when $\inf_{l \geq 1} \{T_l\} > 0$. If not, there exists a subsequence of $\{t_l\}_{l \geq 1}$, namely $\{t_{l_j}\}_{j \geq 1}$ such that $\lim_{j \rightarrow \infty} T_{l_j} = 0$. This implies that there exists $j^* \in \mathbb{R}$ satisfying $j^* = \lim_{j \rightarrow \infty} t_{l_j} < +\infty$, which follows from (35) that

$$\sigma\|\varphi_i(t_l^i)\| + \varepsilon_i(t) \leq \Gamma \cdot 0 = 0.$$

This contradicts with $\sigma\|\varphi_i(t_l^i)\| + \varepsilon_i(t) > 0 (t \geq 0)$. Therefore, Zeno behavior does not occur during the control period. $\square$

4. A numerical example

As a numerical example, we use the isolated INNs (3), where $A = \text{diag}(2.07, 0.37, 0.702)$, $Z = \text{diag}(0.97, 1.2, 2.21)$, $g(\cdot) = \tanh(\cdot)$, $I_{v1} = I_{v2} = I_{v3} = (0, 0, 0)^T$, and

$$F = \begin{pmatrix} 1.35 & -2.12 & 3.29 \\ -2.02 & -0.069 & 2.2 \\ -2.382 & -2.16 & 1.267 \end{pmatrix}.$$

Phase trajectories of the isolated INNs (3) under three distinct initial conditions are plotted in Figs. 1, 2–3.

Consider a digraph $\bar{\mathcal{G}}$ composed of 9 INNs (1) with three leaders, which is illustrated in Fig. 4. Clearly, $\bar{\mathcal{G}}$ is divided into three clusters,

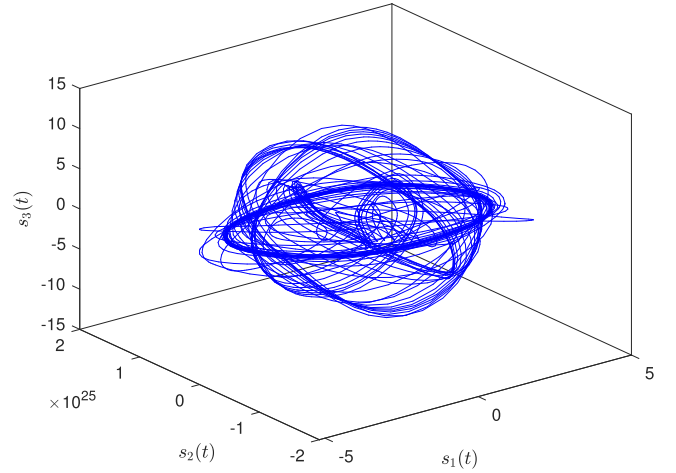


Fig. 3. The portraits of INNs (3) with $S_3(0) = (0.6, -1.3, -1.5)^T$.

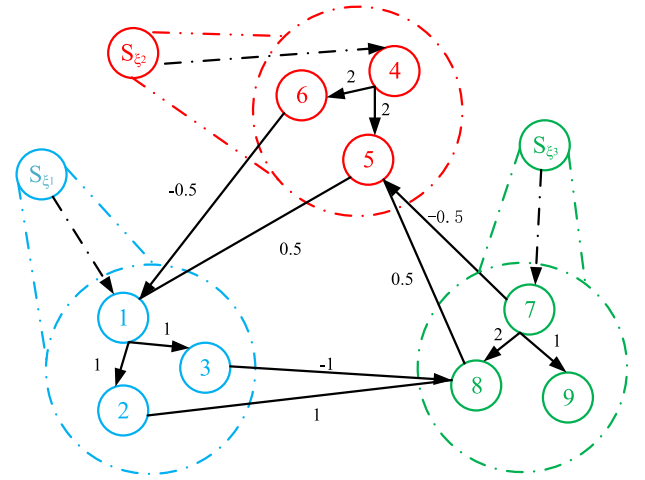


Fig. 4. Directed interaction graph of INNs (1) with $P = 9$.

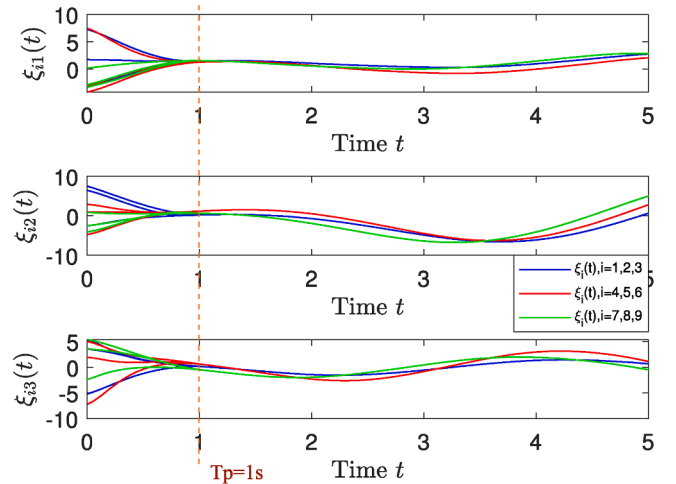


Fig. 5. Trajectories of $\xi_i(t)$ ($i = 1, 2, \dots, 9$) and the corresponding virtual nodes $S_i(t)$ ($i = 1, 2, 3$).

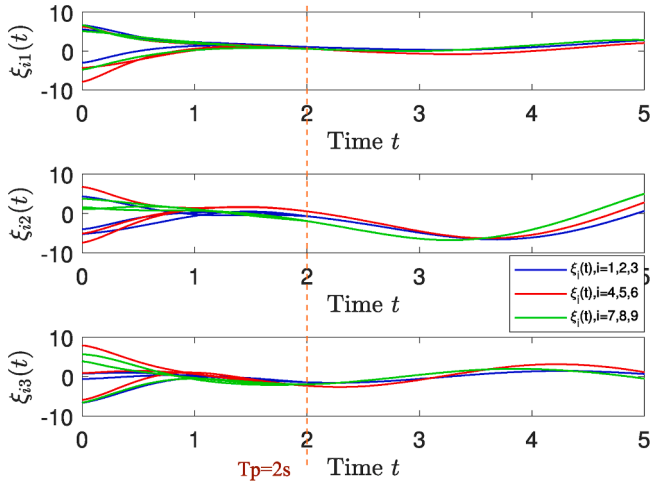


Fig. 6. Trajectories of $\xi_i(t)$ ($i = 1, 2, \dots, 9$) and the corresponding virtual nodes $S_i(t)$ ($i = 1, 2, 3$).

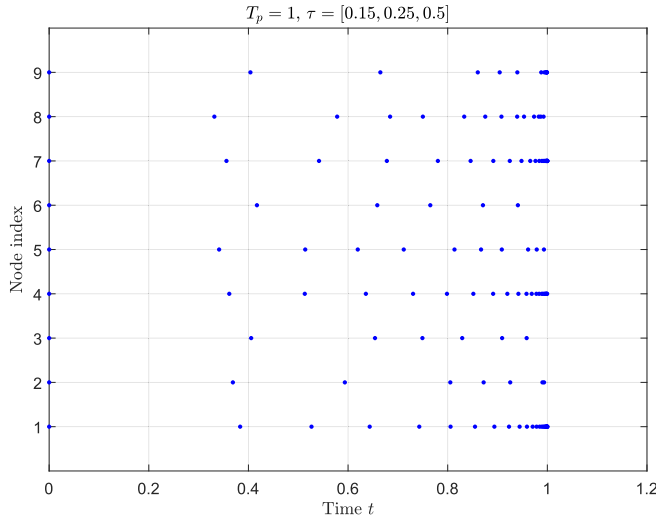


Fig. 7. Event-triggered instants of INNs for $T_p = 1s$.

i.e., $\Delta_1 = \{1, 2, 3\}$, $\Delta_2 = \{4, 5, 6\}$, and $\Delta_3 = \{7, 8, 9\}$, and each cluster is associated with corresponding virtual nodes S_1, S_2 , and S_3 . Moreover, if the pinned nodes are chosen as $Y_1 = \{1\}$, $Y_2 = \{4\}$, $Y_3 = \{7\}$, and $d_1 = d_4 = d_7 = 3$, it is evident that $\bar{G}$ possesses a spanning tree, i.e., Assumption 3 holds. Set $\ell_o = 1$, ($o = 1, 2, 3$), then Assumption 2 holds.

Selecting $\Lambda = I_9$ and $\Lambda_1 = \Lambda_2 = \Lambda_3 = I_3$, one has $\lambda_{\min}(\Lambda L_s + L_s^T \Lambda) = -0.9239$, $\lambda_{\min}(\Lambda_1 \bar{L}_{11} + \bar{L}_{11}^T \Lambda_1) = 3.0140$, $\lambda_{\min}(\Lambda_2 \bar{L}_{22} + \bar{L}_{22}^T \Lambda_2) = 1.0258$, and $\lambda_{\min}(\Lambda_3 \bar{L}_{33} + \bar{L}_{33}^T \Lambda_3) = 1.5842$. Thus $\Lambda_i \bar{L}_{ii} + \bar{L}_{ii}^T \Lambda_i > 0$ ($i = 1, 2, 3$) holds in Lemma 1, and $\Lambda \bar{L} + \bar{L}^T \Lambda > \rho \Lambda$ holds for $\rho = 2$.

Select the coupling strengths as $\omega_1 = 3$, $\omega_2 = 5$, $\omega_3 = 2$, and let $\kappa = 0.4$, then $\theta_1 = 28.8$, $\theta_2 = 3.014 > 0$. This also means that (19) holds. Set $\sigma = 0.5$, $\varsigma_1 = 0.2$, $\varsigma_2 = 0.6$, it is easy to obtain that $w_1 = 31.2195$, $w_2 = -4.0186$, $w_3 = 2.4$ and $w_4 = 5.2186$. Thus, one gets $1 - \frac{w_4}{2\epsilon} = -0.2302 < 0$, where $\epsilon = 2.1211$. According to Theorem 1, the INNs (1) with 9 nodes can realize prescribed-time cluster lag synchronization. Fig. 5 presents the evolution of $\xi_i(t)$ ($i = 1, \dots, 9$) and $S_i(t)$ ($i = 1, 2, 3$) for $T_p = 1$, $\tau_1 = 0.15$, $\tau_2 = 0.25$, and $\tau_3 = 0.5$. Fig. 6 displays the trajectories of $\xi_i(t)$ ($i = 1, \dots, 9$) and $S_i(t)$ ($i = 1, 2, 3$) for $T_p = 2$, $\tau_1 = 0.2$, $\tau_2 = 0.4$, $\tau_3 = 0.7$.

It can be observed from Figs. 5 and 6 that the proposed scheme achieves predefined-time cluster lag synchronization effectively for different prescribed times T_p , with the actual synchronization time being no later than the designated value, thereby validating the theoretical results.

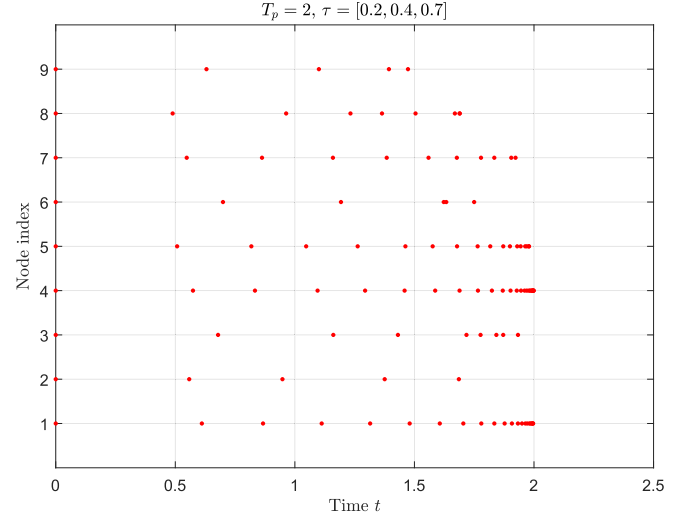


Fig. 8. Event-triggered instants of INNs for $T_p = 2s$.

To further validate the effectiveness of the dynamic event-triggered mechanism, Figs. 7 and 8 display the event-triggered instants for each node corresponding to the scenarios in Fig. 5 and Fig. 6, respectively. The discrete and well-distributed triggering instants confirm that Zeno behavior is excluded and illustrate that control updates occur only when necessary, thereby reducing communication while achieving predefined-time cluster lag synchronization.

5. Conclusions

This paper has addressed the predefined-time cluster lag synchronization of INNs. Sufficient conditions for guaranteeing synchronizing INNs in this predefined-time cluster lag manner. have been obtained using dynamic event-triggered control. Furthermore, Zeno behavior has been precluded during the control process. Considering practical challenges in engineering applications, such as time-varying coupling strengths, time-varying delays, and denial-of-service attacks, future work may focus on extending the proposed method using techniques including intermittent event-triggered control or resilient control to achieve synchronization under these conditions.

CRediT authorship contribution statement

Peng Liu: Methodology; Yiwei Shao: Writing – original draft; Yin Sheng: Software; Qiang Xiao: Supervision; Jian Yong: Writing – review & editing.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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